Shravan Asati

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Education

Adani University *July 2023 – July 2027*

B.Tech. in Computer Science and Engineering (CGPA: **9.2/10**)

- **Coursework:** Machine Learning, Distributed Computing, Cybersecurity, Cloud Computing

Experience

Software Engineer Intern *Remote*

Polarcut *Feb 2026 - March 2026*

- Engineered a Remotion-based caption system with frame-accurate timing, word-level motion effects, and LLM-driven semantic emphasis across 4 stylized templates, reducing subtitle production time by 70% per video.
- Implemented an internal AI prompt-to-motion-graphics pipeline for text and vector animations powering Shorts B-roll, replacing a third-party API and reducing estimated per-asset inference cost from INR60 to INR12 (80% lower cost).

Software Engineer Intern *Remote*

Toucan Labs *June 2025 – Aug 2025*

- Architected and deployed a full-stack, enterprise-grade observability platform using Grafana Cloud, leveraging OpenTelemetry standards to provide unified logging (Loki), metrics (Prometheus/Mimir), and Application Performance Monitoring (APM) with distributed tracing (Tempo).
- Designed and implemented a centralized notifications system, including an in-app notification center, Firebase push notifications, and an admin announcement panel, delivering real-time updates with 98%+ delivery success.
- Deployed and managed the full production stack (frontend, backend, MongoDB) on a VPS using Dokploy, with rolling blue-green deployments and a GitHub Actions CI/CD pipeline, enabling fully-automated zero-downtime releases.

Skills

Languages and Frameworks: Python, Go, Next.js, React, TypeScript, JavaScript, Java, SQL, C++, Bash

Tools: Git, GitHub, Linux, Docker, DigitalOcean, Azure, CI/CD Pipelines, NATS, Protocol Buffers

Projects

reverie Website YouTube

- Built a full-stack AI journaling platform serving 50+ users, with a Next.js frontend on Vercel, Spring Boot REST API, and a FastAPI NLP microservice for real-time sentiment and emotion analysis deployed on Azure Container Apps.
- Engineered a RAG pipeline with an NLU engine that extracts filters from user queries, performs hybrid SQL + vector search via pgvector, and applies knee-point detection on embedding distance scores to dynamically threshold retrieval before prompting the LLM.
- Routed AI inference through Cloudflare AI Gateway across multiple models (BGE-M3, GPT-OSS), with end-to-end observability via Sentry across every service layer.

shadowfax *GitHub*

- Built a minimal HTTP/1.1 server from scratch in Go over raw TCP - implemented request parsing, routing, chunked-encoding, persistent connections, middleware, panic recovery, graceful shutdown, and concurrent request handling.
- Developed middleware for CORS, CSRF protection, static file serving, basic authentication, and request logging.
- Achieved **1.4k req/s** under resource-constrained conditions, and ensure robustness through unit and integration testing.

redline *GitHub*

- Designed and built a distributed monitoring architecture with a timing-wheel scheduler, NATS-based task dispatch, stateless regional workers, and an asynchronous alerting pipeline.
- Benchmarked 10k synthetic monitors, sustaining 65 probes/sec with **5% dispatcher CPU and 47 MB memory usage**.
- Implemented worker discovery using NATS KV heartbeats and message deduplication, with durable monitor configuration enabling dispatcher state reconstruction after restarts.

Leadership and Achievements

Vice President, Adani Student Programming and Development Club *Aug 2024 – July 2025*

- Organized technical events such as hackathons, coding challenges, and workshops, impacting over 200 students.

Winner, Techathon 2025 *Jan 2025*

- Awarded first place for building a website focused on road safety and accident prevention.

Second Runner Up, Odoo Hackathon 2026 *Jan 2026*

- Secured second runner up place for building a restaurant POS system, streamlining retail transaction workflows.